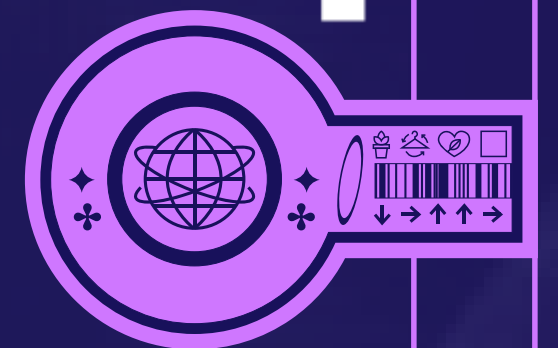


RISK MANAGEMENT OF LARGE CONSTRUCTION PROJECTS

BRING OUT THE MARITES IN YOU!

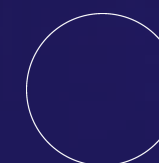




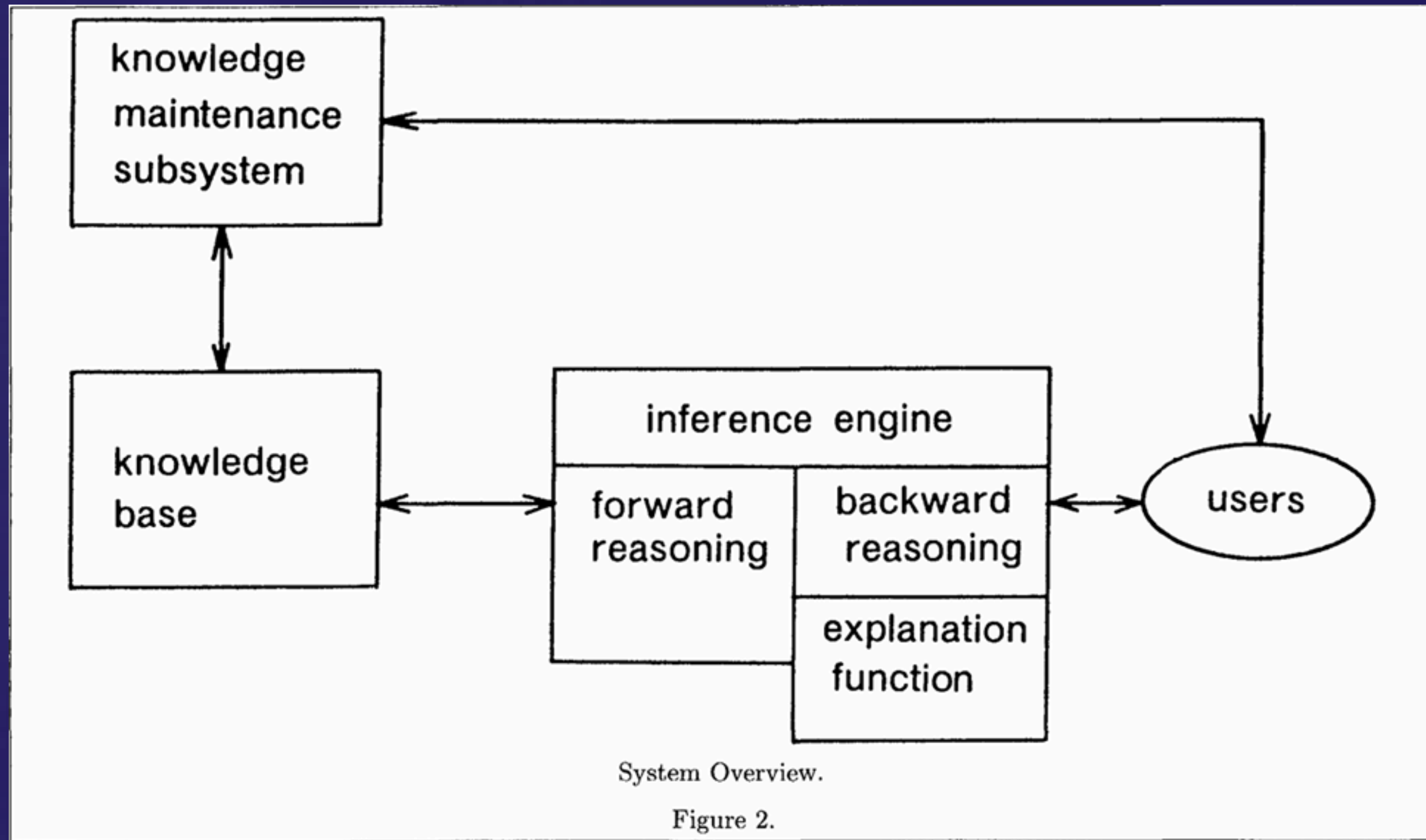
◀ ROUND 1 ▶



OVERVIEW



The study compares four knowledge representation schemes—simple production system, structured production system, frame system, and logic system—through pilot expert systems for risk management in large construction projects. It assesses ease of implementation and run-time efficiency, using the same domain knowledge and functionality. Findings show modular systems (simple production, logic) are easier to implement but less efficient, particularly with larger knowledge bases. Structured systems (structured production, frame) are harder to build but provide better inference speed and scalability. The research highlights key trade-offs between implementation complexity and performance, offering practical guidance for expert system design.



THE PROBLEM



Risk management of large construction projects

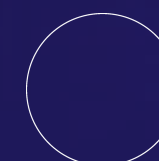
Large-scale construction projects face various risks during execution, which can cause delays, cost overruns, and technical issues. Project managers must continuously monitor and manage these risks to ensure success. However, the complexity and dynamic nature of such projects make risk identification, prediction, and response challenging.



« ROUND 2 »



OBJECTIVES



OBJECTIVES

01 To help project managers effectively control their projects.

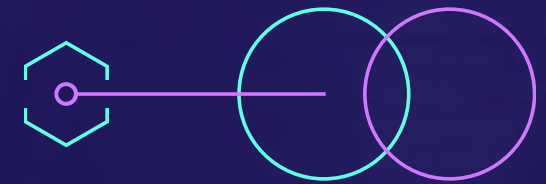
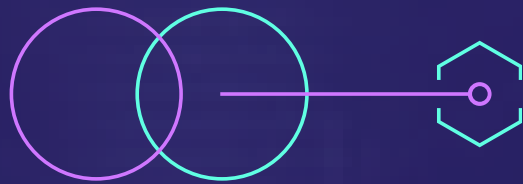
02 To provide them with appropriate and relevant knowledge.

03 To gather knowledge from many expert project managers

04 To integrate the collected knowledge into a centralized knowledge base.

STEPS OF SOLUTION

《 ROUND 3 》



We propose developing an expert system using the Simple Production System approach, implemented in CLIPS, to assist project managers in managing risks in large construction projects. This solution will



- Represent knowledge using clear IF-THEN rules



- Allow forward reasoning to predict potential risks based on input causes



⏮️ ROUND 4 ⏭️

WHY CHOOSE A SIMPLE PRODUCTION SYSTEM?



KEY REASONS:

-
- 01 Easiest to Implement - Simple "IF-THEN" rules, minimal setup
- 02 Faster Development - No complex hierarchies or structures needed
- 03 Transparent & Debuggable - Rules are human-readable and easy to modify
- 04 Good for Small/Medium Problems - Efficient for well-defined domains

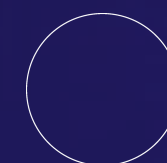


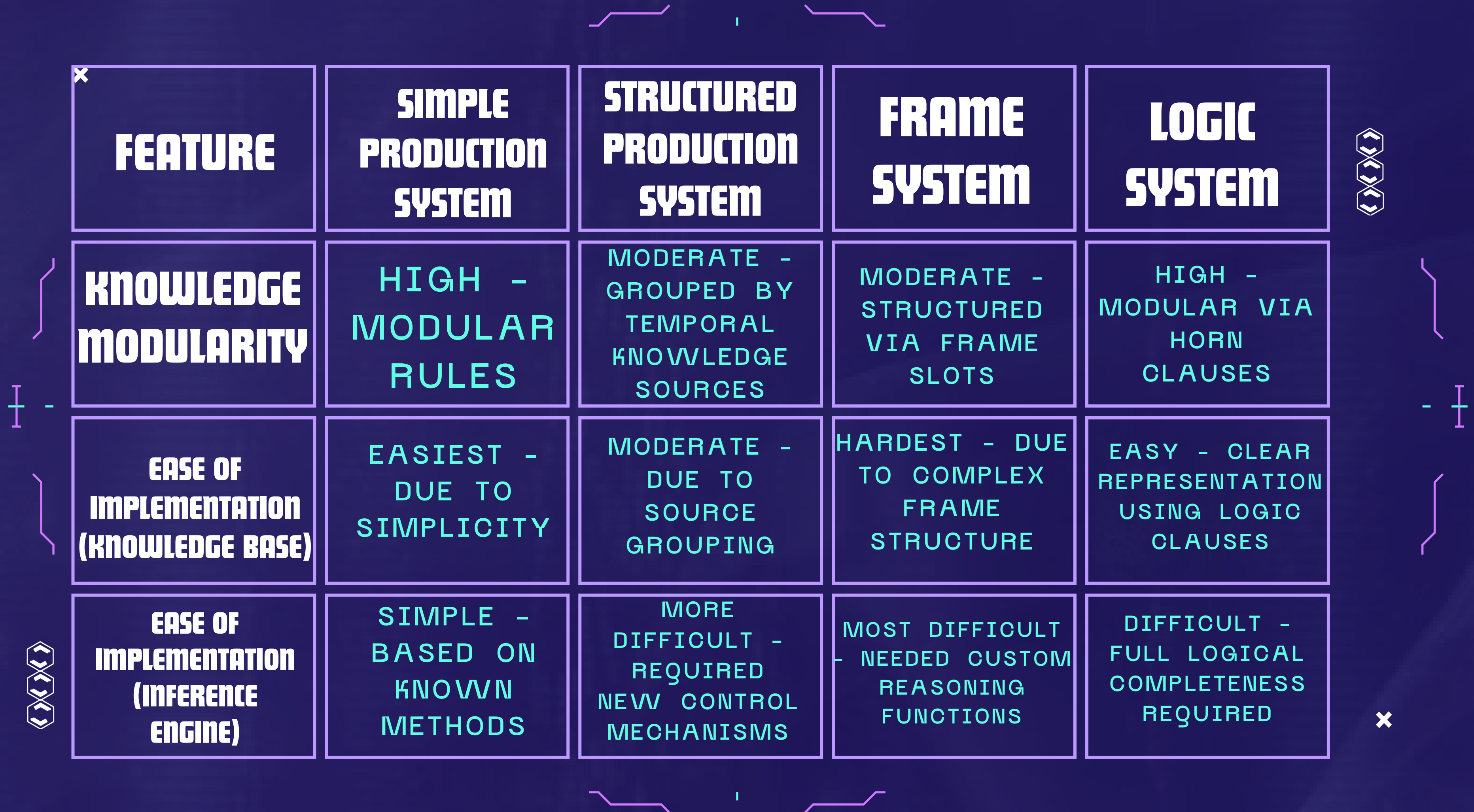


◀ ROUND 4 ▶



CHARACTERISTICS
> COMMON TO ALL <
FOUR SYSTEMS

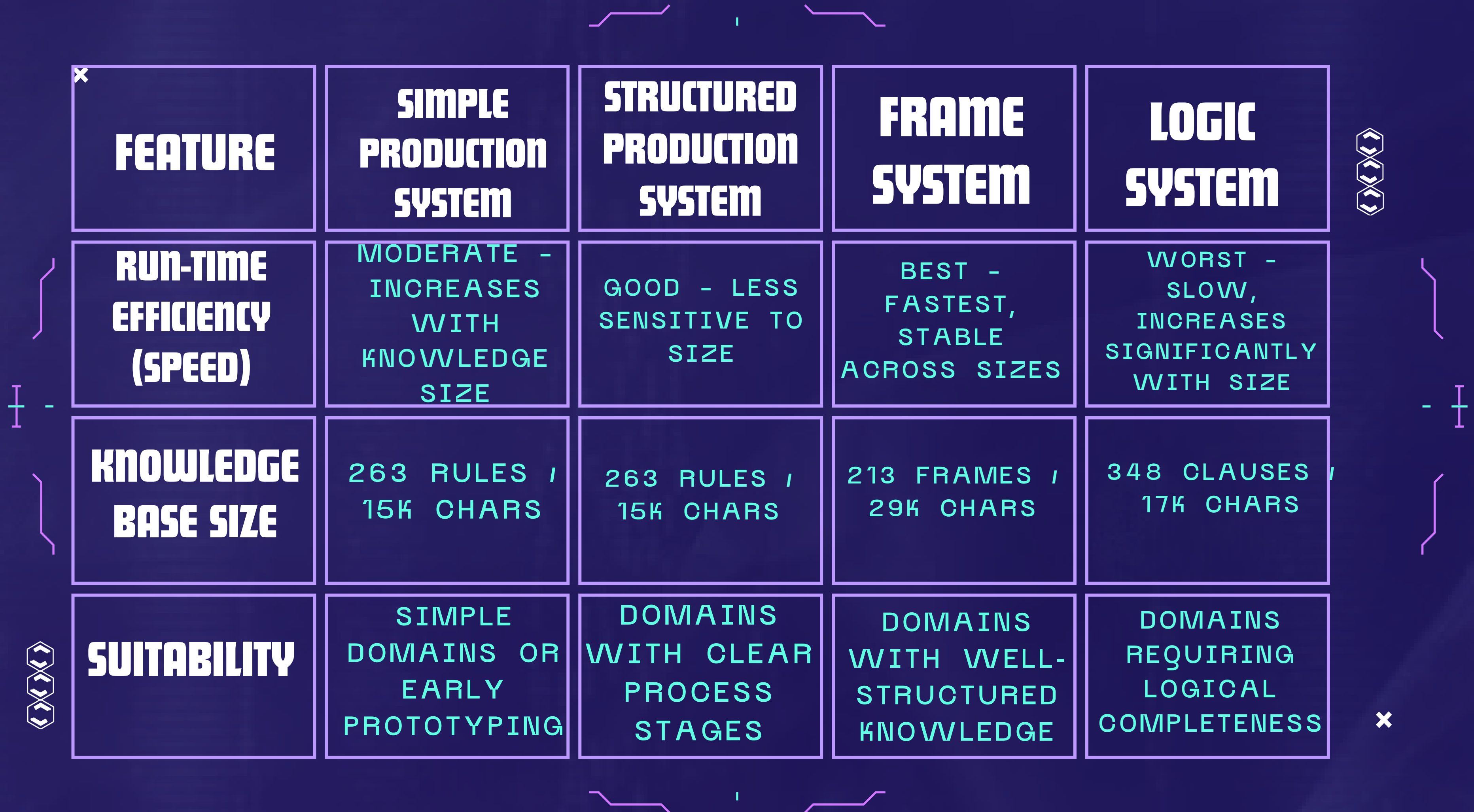




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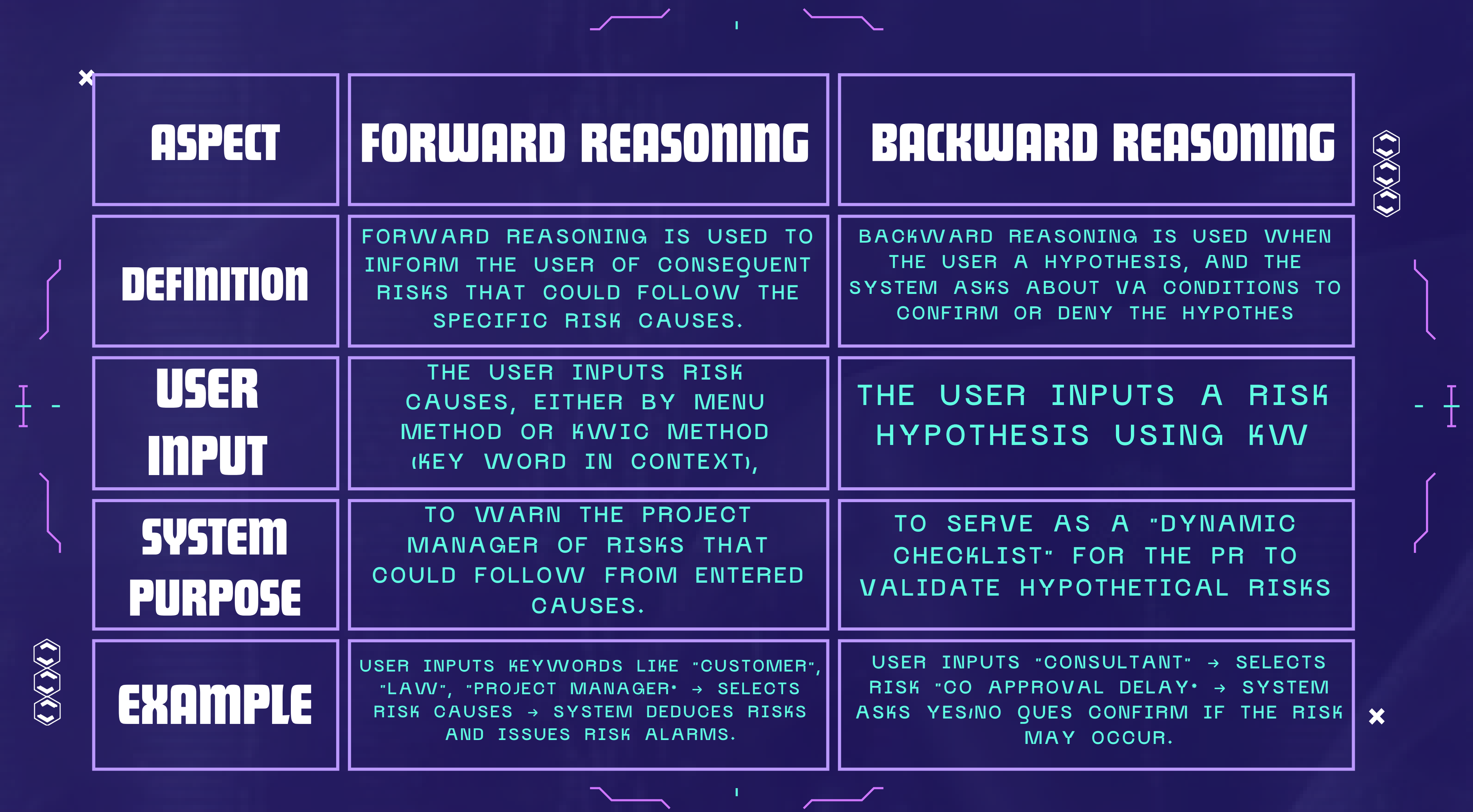
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COMPARISON
OF FORWARD
AND BACKWARD
CHAINING

« ROUND 5 »





《 ROUND 6 》

REPRESENTING KNOWLEDGE



01

The most common reason that a project faces approval delays is a lack of customer or consultant ability and the project manager's misguidance.

02

The most common reason that a project requires spare parts re-orders is because of a contractual defect in the scope of equipment supply.

03

Spare parts air cargo due to incomplete delivery often indicates a misunderstanding of the spare parts amount to be delivered; to address this, review delivery specifications thoroughly.

04

The most common reason that a project requires additional equipment is because of a lack of thorough examination of all specifications by project managers.





05

The most common reason that a project requires a pipe foundation change for big equipment carry-in is because of a contractual defect in material standards.

06

The most common reason that a project faces material retests is because of poor interpretation by the inspection company due to complicated laws or those different from Japan's.

07

The most common reason that a project faces special tools excessive requests is because of incomplete confirmation stemming from a contractual defect in the scope of equipment supply

08

The reason that a project faces excessive delivery is because of no revision of delivery scope after design change, caused by a contractual defect in the scope of equipment supply





09

The most common reason that a project faces a customer's rejection of project acceptance is because of a misunderstanding of the spare parts amount to be delivered.

10

The most common reason that a project faces a material upgrade request for the customer's future plan is because of a contractual defect in the scope of equipment supply and approval activities.

11

The most common reason that a project faces a contractual defect in technical guarantee is because of complicated laws or those different from Japan's and a lack of examination by project managers.



—12

The most common reason that a project faces a civil work delay is because of bad negotiation with the consultant due to a lack in customer or consultant ability.





13

The most common reason that a project faces a sudden material change is because of a consultant error stemming from a lack in customer or consultant ability.

14

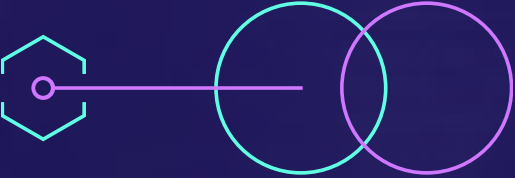
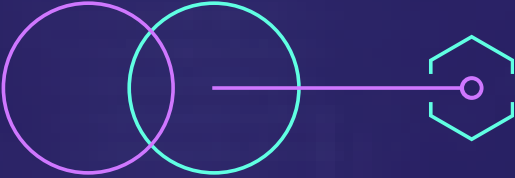
The most common reason that a project faces a decrease in work efficiency is because of a contractual defect in the scope of equipment supply.



« ROUND 7 »

REPRESENTING KNOWLEDGE

(FORWARD REASONING - IF/THEN RULES)



IF

THEN

THERE IS A LACK IN CUSTOMER
OR CONSULTANT ABILITY AND
PROJECT MANAGER MISGUIDANCE

THE PROJECT WILL LIKELY FACE
APPROVAL DELAYS

THERE IS A CONTRACTUAL DEFECT
IN THE SCOPE OF EQUIPMENT
SUPPLY

THE PROJECT WILL LIKELY
REQUIRE SPARE PARTS RE-ORDERS

SPARE PARTS ARE DELIVERED VIA
AIR CARGO DUE TO INCOMPLETE
DELIVERY

INDICATES A MISUNDERSTANDING
OF THE SPARE PARTS AMOUNT;
REVIEW DELIVERY SPECIFICATIONS

THERE IS A CONTRACTUAL DEFECT
IN THE SCOPE OF EQUIPMENT
SUPPLY

THE PROJECT MAY FACE A
DECREASE IN WORK EFFICIENCY

PROJECT MANAGERS FAIL TO EXAMINE ALL SPECIFICATIONS THOROUGHLY	THE PROJECT WILL LIKELY REQUIRE ADDITIONAL EQUIPMENT
THERE IS A CONTRACTUAL DEFECT IN MATERIAL STANDARDS	THE PROJECT WILL LIKELY REQUIRE A PIPE FOUNDATION CHANGE FOR BIG EQUIPMENT CARRY-IN
THE INSPECTION COMPANY MISINTERPRETS LAWS DUE TO THEIR COMPLEXITY OR DIFFERENCES FROM JAPAN'S	THE PROJECT WILL LIKELY FACE MATERIAL RETESTS
THERE IS A CONTRACTUAL DEFECT IN EQUIPMENT SCOPE AND CONFIRMATION IS INCOMPLETE	THE PROJECT WILL LIKELY FACE EXCESSIVE SPECIAL TOOLS REQUESTS
THE DELIVERY SCOPE IS NOT REVISED AFTER A DESIGN CHANGE, AND THERE IS A CONTRACTUAL DEFECT IN EQUIPMENT	THE PROJECT WILL LIKELY FACE EXCESSIVE DELIVERY

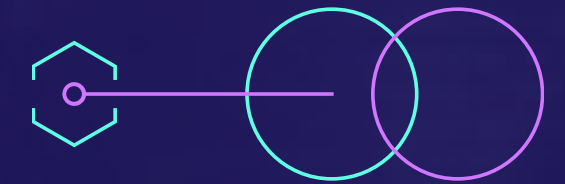
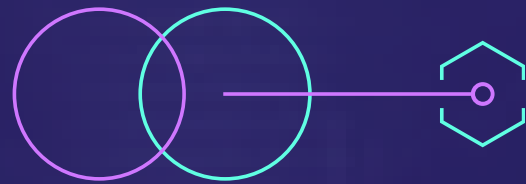
THERE IS A MISUNDERSTANDING OF THE SPARE PARTS AMOUNT TO BE DELIVERED	THE CUSTOMER MAY REJECT PROJECT ACCEPTANCE
THERE IS A CONTRACTUAL DEFECT IN BOTH SCOPE OF SUPPLY AND APPROVAL ACTIVITIES	THE CUSTOMER MAY REQUEST A MATERIAL UPGRADE FOR FUTURE PLANS
LAWS ARE COMPLICATED OR DIFFER FROM JAPAN'S AND PROJECT MANAGERS FAIL TO EXAMINE THEM	THE PROJECT WILL LIKELY FACE A CONTRACTUAL DEFECT IN TECHNICAL GUARANTEE
THE CUSTOMER OR CONSULTANT LACKS ABILITY AND CONSULTANT NEGOTIATION IS POOR	THE PROJECT WILL LIKELY FACE CIVIL WORK DELAY
THE CONSULTANT MAKES AN ERROR DUE TO LACK IN CUSTOMER OR CONSULTANT ABILITY	THE PROJECT MAY FACE A SUDDEN MATERIAL CHANGE





« ROUND 8 »

FACTS





01

There is a lack in customer or consultant ability and project manager misguidance.

02

There is a contractual defect in the scope of equipment supply.

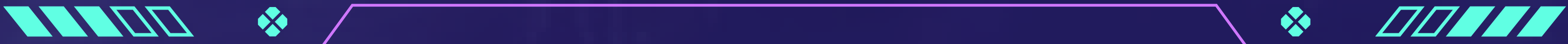
03

Spare parts are delivered via air cargo due to incomplete delivery.

04

Project managers fail to examine all specifications thoroughly.





05 There is a contractual defect in material standards.

06 The inspection company misinterprets laws due to their complexity or differences from Japan's.

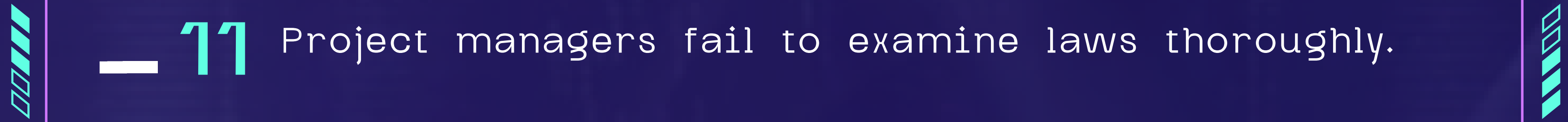
07 Confirmation is incomplete.

08 The delivery scope is not revised after a design change.

09 There is a misunderstanding of the spare parts amount to be delivered.

10 There is a contractual defect in both the scope of equipment supply and approval activities.

11 Project managers fail to examine laws thoroughly.





12 Negotiation with the consultant is poor.

13 The consultant makes an error



please key-in strings for risk causes or?

>customer, law, project manager

*please select risk cause codes

1g05 contractual defect in time for approval
3O01 lack in customer english ability
3Q01 lack in customer or consultant ability
3Q03 different business practices of customers
3k01 complicated laws or those different from Japan's
3k02 law or regulation change
2e31 project manager misguidance
2g31 lack in examination of project managers

>3k01, 3Q03, 2g31

*please select activity codes or all

>all

rule 1016 deduced(1d02 contractual defect in technical
guarantee)
deduced(1f01 contractual defect in material
standard)

rule 3018 deduced(6105011 spare parts air cargo due to
incomplete delivery)

risk alarm

2103002 approval delay due to misguiding spare parts
amount

2103011 civil approval delay due to loading data
between civil and equipment differences

2202013 Additional equipment because no one
examines all specifications thoroughly

2202012 spare parts re-order due to number disorder
5103013 pipe foundation change for big equipment
carry-in

2304003 material re-test due to inspection company
poor interpretation

6105011 spare parts air cargo due to incomplete
delivery



OUR TEAM

- Habiba Eteam
- Rahma Yahia
- Rahma Hassan
- Basma Ali
- Malak Ehab
- Yosr Wael



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